TrialHub CTMS — Current-State Architecture Assessment

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1. Executive summary

TrialHub is Vernova's in-house Clinical Trial Management System (CTMS), used by Clinical Operations to plan and run approximately 120 active clinical studies across 25 countries. The platform was commissioned in 2011 as a Java EE monolith and has not been materially re-architected since. It is a Good Clinical Practice (GCP) critical system subject to 21 CFR Part 11, EU GMP Annex 11, and GDPR (special-category subject data). Every core technology layer is now past vendor end-of-support.

TrialHub supports ~1,800 users: Clinical Operations staff, Clinical Research Associates (CRAs), and investigator-site users. It integrates with the Electronic Data Capture (EDC) system via SOAP, the IRT/randomisation vendor via SFTP, the safety database via a nightly ETL, and SharePoint TMF. The release cadence is currently 6 months — driven almost entirely by the need to re-validate the full EAR for any change.

The system is on the FY27 modernization shortlist. A European Medicines Agency inspection in Q4 FY26 raised an observation about the 25-year TMF archive readability drill not having been performed since 2023; this has elevated the modernization priority.

Headline numbers - Users: ~1,800 across 25 countries. - Active studies: ~120. - Database size: ~1.2 TB (Oracle 11g R2). - Code base: ~180,000 lines of Java 7 across a single EAR. - Release cadence: 6 months. - Annual run cost (hosting + Oracle + WebLogic licences + support): approximately €1.4M.

2. Business context

TrialHub is the system of record for clinical trial operations: - Study planning, protocols, and amendments - Investigator-site management and qualification - Subject enrolment milestone tracking - Monitoring visit planning, reports, and action items - Trial Master File (TMF) document references (actual files live in SharePoint) - Inspection readiness evidence

RTO: 8 hours. RPO: 24 hours.

Regulatory scope: - **FDA 21 CFR Part 11** — electronic records / e-signatures - **EU GMP Annex 11** — computerised systems - **ICH E6(R2) GCP** — Good Clinical Practice guidelines - **GDPR** — pseudonymised subject data is Article 9 special category - **GAMP 5** — Category 5 (custom-built) - **TMF Reference Model** — document taxonomy

3. Logical architecture

Three-tier Java EE monolith: 1. **Presentation**: JSF 1.2 + RichFaces 3, server-rendered. IE 11 compatibility mode still required for ~30% of users. 2. **Business logic**: EJB 3.0 stateless session beans, Hibernate 3 for persistence. ~180k LOC in a single EAR deployed on WebLogic. 3. **Data**: Oracle Database 11g R2 Enterprise. Heavy use of PL/SQL packages for business logic — approximately 85 packages, 42 of which are tagged as holding validated clinical rules (randomisation, subject-status transitions, visit-window calculations).

A separate Quartz-based scheduler runs inside the EAR for nightly batch ETL to the safety database.

4. Physical architecture

On-premises deployment at DC-EU-DUB (primary) and DC-US-RDU-COLO (DR):

Role	Host	Spec
WebLogic cluster	4-node (web001–web004)	16 vCPU, 64 GB RAM each, RHEL 6
Oracle database	2-node Oracle RAC	32 vCPU, 256 GB RAM, SAN 2 TB
Admin / batch node	1 node (adm001)	8 vCPU, 32 GB RAM
HSM for e-signatures	2 x Thales nShield 500	hardware

APAC sites hit DC-EU-DUB over the corporate WAN with ~250 ms RTT, causing complaints.

5. Technology stack

Component	Version	Vendor support status
Java	7 (OpenJDK)	**End of life** (Oracle premier support ended 2015, extended 2022)
WebLogic Server	10.3.6	**End of life** (ended Dec 2021)
Oracle Database Enterprise	11g R2	**End of life** (extended support ended Dec 2020)
Red Hat Enterprise Linux	6	**End of life** (maintenance support ended June 2024)
JSF	1.2	**Obsolete** (Jakarta EE moved to Faces 4.x; namespace change javax → jakarta required)
RichFaces	3.3.4	**End of life** (project retired in 2016)
Hibernate	3.6	Several majors behind (current 6.x)
Apache Ant	1.8	Legacy build tool, no dependency management
Quartz Scheduler	1.8	Several majors behind (current 2.3)

Software composition risk: The Information Security team's SCA scan from July 2026 flagged 47 critical and 118 high CVEs across these dependencies.

6. Integrations

6.1 EDC (Medidata Rave) — bi-directional, SOAP

Subject enrolment and visit data flows in; TrialHub posts site-status changes out. The SOAP contract is tightly coupled to EJB method signatures — a Rave API change in Q2 FY26 required a 3-week emergency patch.

6.2 IRT/Randomisation vendor — inbound, SFTP

Daily batch files land in a landing directory for Quartz to pick up. No cryptographic verification of file contents. This is an open data integrity observation (GCP-OBS-2025-033) — fails ALCOA+ "Original" criterion.

6.3 Safety database (Argus Safety) — outbound, ETL

Nightly Quartz job extracts safety-relevant data and loads into Argus via DB-link. Failures surface only in Quartz logs — no monitoring, no retries. Has silently dropped data on three occasions in the past 18 months.

6.4 SharePoint TMF — outbound, document links

TrialHub stores URLs to TMF documents in SharePoint. Actual document content is in SharePoint, not TrialHub.

6.5 LDAP authentication

Users authenticate via corporate LDAP (identical to LabTrack LIMS setup). External site users managed manually by Clinical Operations in a parallel Excel workbook — a long-standing audit finding.

7. Non-functional characteristics

Performance: EU/US sites acceptable. APAC sites report page response > 15 s during peak enrolment windows. Root cause: server-rendered JSF requires many round-trips over long-haul WAN.

Availability: 99.1% over the past 12 months. One extended outage of 14 hours in March 2026 when the Oracle cluster failed over and the WebLogic connection pool did not recover automatically.

Scalability: Stateful JSF sessions bind each user to a specific cluster node. Scaling means buying more RAM for existing nodes — the WebLogic cluster hasn't horizontally scaled since 2017.

DR: DC-US-RDU-COLO holds a cold standby. Last successful DR failover exercise: September 2024. The February 2026 exercise failed when the DR Oracle backup proved to be 72 hours stale.

8. Operations and support

- **Deployment:** Full EAR redeployment every 6 months. Downtime window: 8 hours.
- **Change control:** CAB + QA approval. Average lead time: 12 to 18 weeks for a non-trivial change.
- **Testing:** No automated tests. OQ regression suite is ~450 scripts executed by 3 QA analysts over 10 working days per release.
- **Monitoring:** Nagios on host metrics only. No APM. Application logs rotate on local disk, kept 30 days.
- **Backup:** Oracle RMAN nightly backups to local disk, swept to tape weekly, 25-year retention per TMF Reference Model.

9. Compliance posture

Validation status

Validated under GAMP 5 Category 5 (custom). Current validation documents (URS, FS, DS, RA) last amended March 2025.

21 CFR Part 11 implementation

- **Audit trail:** Implemented in-database on 63 tables. Includes 25-year retention.
- **E-signatures:** HSM-backed (Thales nShield). Used for protocol amendment approval, monitoring visit report sign-off, subject-status change authorisations.

- **MFA:** Not enforced on signature events — password re-entry only. Open internal observation GXP-OBS-2024-091.

GDPR implementation

Subject identifiers are pseudonymised in the application layer before persistence. The pseudonymisation key store lives on the WebLogic admin node — a single point of failure that was flagged in the Q4 FY25 DPIA review.

Open audit findings

ID	Severity	Open since	Description
GCP-OBS-2024-091	Medium	Sep 2024	No MFA on e-signature events
GCP-OBS-2025-033	Medium	Apr 2025	IRT files not cryptographically verified (ALCOA+ Original)
GCP-OBS-2025-061	High	Aug 2025	WebLogic/RHEL6 end-of-life, no patches
EMA-OBS-2026-007	High	Oct 2026	25-year TMF archive readability drill not performed since 2023

10. Pain points and risks

- **QA lead:** "Six-month releases are not acceptable to our Clinical Operations stakeholders. Any plan that doesn't change the release cadence won't be funded."
- **Lead developer (R. Shankar):** "The visit-window PL/SQL is the most dangerous thing in this system. If we refactor it wrong, we'll miss or trigger subject safety visits incorrectly."
- **Clinical Operations head:** "APAC sites are losing productivity. If performance isn't fixed by H1 FY28 we'll have to procure a separate regional instance."
- **Information Security:** "The HSM is end-of-life in 2028. If we're still on-premises then, we need a hardware refresh plan now."

11. Prior modernization attempts

- **2017:** A planned WebLogic 12c upgrade was descoped when the JSF 1.2 → 2.3 migration proved to require rewriting ~40% of UI code.
- **2024:** A proof-of-concept to extract subject-enrolment as a Spring Boot microservice reached a working MVP but was shelved when the business owner moved to another division.

12. Constraints affecting modernization

- Validated state preservation — risk-based CSA plan is a prerequisite.
- No cutover during a study database lock (currently 2 locks scheduled Q1 FY28; 4 in Q2 FY28).

- Subject data for EU trials must remain in the EU.
- Audit trail continuity must be demonstrable to inspectors.
- Budget envelope: €3.5M–€4.5M over FY27–FY29.
- The HSM is end-of-life in 2028 — the modernization plan must address e-signature infrastructure.

13. Expected outcomes

- Shorten release cadence from 6 months to monthly.
- Reduce APAC page response below 3 seconds.
- Replace the end-of-life HSM with cloud-managed signing.
- Automate the OQ regression suite.
- Enable self-service onboarding for investigator-site users with MFA.
- Restore the 25-year TMF archive drill to annual cadence.